

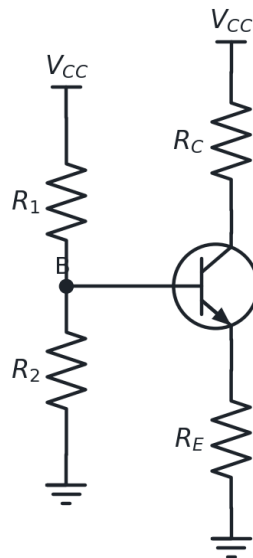
# EDC §1.6 — Voltage-Divider Bias: the Malvino Method

Rewrite of Chapter-1 §1.6 per class guidelines (Malvino): stiff/firm via  $R_{Th} \cdot V_{CC}$ -dependent design rules · ENEX 151 / EX 151

**What changed vs the earlier Chapter-1 notes:** this rewrite replaces the Floyd-Boylestad " $R_2 = \beta R_E / 10$ " table with **Malvino's method** as taught in your class — stiff/firm defined on  $R_{Th}$ , and design guidelines that **depend on the size of  $V_{CC}$** . Everything is derived, not just stated. In the exam, name the convention you use ("stiff per Malvino:  $R_{Th} \leq 0.01\beta R_E$ ") — the two textbooks' words overlap but their numbers differ.

## 1.6 Voltage-Divider Bias — the Malvino Method HOT HOT (GUARANTEED QUESTION — 4/4 YEARS)

### A. The circuit, and the one idea behind it



Voltage-divider bias:  $R_1$ - $R_2$  fix the base voltage,  $R_E$  turns it into a fixed emitter current,  $R_C$  develops the output. (For a CC/emitter-follower design: delete  $R_C$ , collector straight to  $V_{CC}$ .)

$R_1$ - $R_2$  hold the base at a fixed voltage;  $R_E$  converts that fixed voltage into a fixed emitter current. That is the whole design.

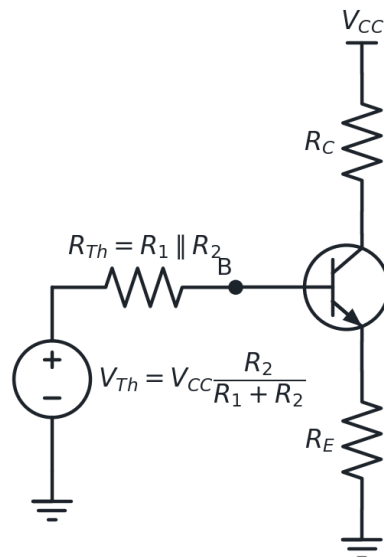
**Physical stabilizing story (worth 1-2 marks, tell it as a chain):** suppose  $I_C$  tries to rise ( $\beta$  higher on this unit, or temperature up). Then  $I_E$  rises  $\rightarrow V_E = I_E \cdot R_E$  rises  $\rightarrow$  since  $V_B$  is held constant by the divider,  $V_{BE} = V_B - V_E$  falls  $\rightarrow$  falling  $V_{BE}$  means the EBJ **decreases the injected charge carriers**, pulling  $I_C$  back down  $\rightarrow I_C$  remains (almost) constant. This is **negative feedback via  $R_E$** :  $R_E$  senses the output current and subtracts from the input voltage.

**KEY** The divider can only "hold  $V_B$  constant" if the base doesn't load it down. Quantifying *how well it holds* is exactly the stiff/firm question below.

### B. Exact analysis — Thévenize the base network

Looking from the base, replace the  $R_1$ - $R_2$  divider by its Thévenin equivalent (open-circuit voltage and dead-source resistance):

$$V_{Th} = V_{CC} \cdot R_2 / (R_1 + R_2) \text{ and } R_{Th} = R_1 \parallel R_2$$



Thévenin equivalent of the base network: one source  $V_{Th}$  behind one resistor  $R_{Th}$  — the whole bias problem becomes a single KVL loop.

**Base-loop KVL** (through  $V_{Th} \rightarrow R_{Th} \rightarrow V_{BE} \rightarrow R_E \rightarrow \text{ground}$ ), with  $I_B = I_E/(\beta+1)$ :

$$V_{Th} = I_B \cdot R_{Th} + V_{BE} + I_E \cdot R_E = I_E \cdot [R_{Th}/(\beta+1)] + V_{BE} + I_E \cdot R_E$$

$$I_E = (V_{Th} - V_{BE}) / [R_E + R_{Th}/(\beta+1)] \text{ — the exact result. } \checkmark$$

**The ideal we are chasing:** if the term  $R_{Th}/(\beta+1)$  were zero,

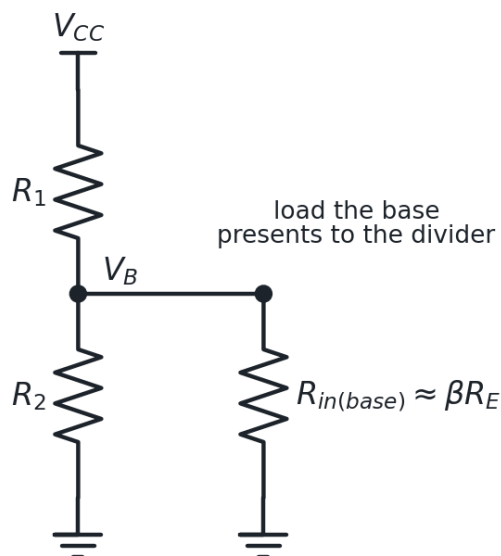
$$I_E(\text{ideal}) = (V_{Th} - V_{BE})/R_E \text{ — no } \beta \text{ anywhere.}$$

So the *entire*  $\beta$ -dependence of the circuit lives in one nuisance term,  $R_{Th}/(\beta+1)$ , sitting next to  $R_E$ . Bias design = making that term negligible.

### C. Stiff and firm biasing — Malvino's rules, derived

**Step 1 — what load does the base put on the divider?** Looking into the base (DC), the emitter resistor is magnified by the reflection rule:

$$R_{in}(\text{base}) = (\beta+1) \cdot R_E \approx \beta R_E$$



What the divider actually drives: the base presents  $R_{in}(\text{base}) \approx \beta R_E$ . The divider is a voltage source with internal resistance  $R_{Th}$  feeding this load.

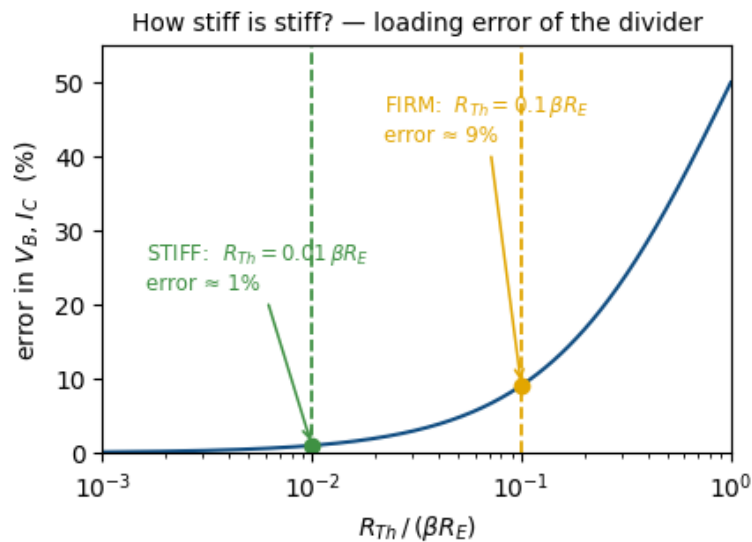
**Step 2 — Malvino's stiff-source idea.** A voltage source is "stiff" for a given load when its internal resistance is at most **1/100 of the load** — then the load steals less than 1% of the voltage. Here the source is the divider (internal resistance  $R_{Th}$ ) and the load is  $\beta R_E$ . The fractional error it causes — in  $V_B$ , and therefore in  $I_E$  and  $I_C$  — is:

$$\text{error} \approx R_{Th} / (\beta R_E) \text{ (exactly: } R_{Th} / (R_{Th} + \beta R_E) \text{ — same thing when small)}$$

(Same conclusion from the exact formula:  $I_E$  is low by the factor  $[R_{Th}/(\beta+1)]/R_E \approx R_{Th}/\beta R_E$  compared with the ideal.)

**Step 3 — the two named design points:**

| Method           | Criterion (Malvino)                | Equivalent form                     | Q-point error                                                     | When to use                                 |
|------------------|------------------------------------|-------------------------------------|-------------------------------------------------------------------|---------------------------------------------|
| a) Stiff biasing | $R_{Th} \leq 0.01 \cdot \beta R_E$ | $R_E \geq 100 \cdot R_{Th} / \beta$ | $\leq \sim 1\%$ — $V_B$ essentially independent of the transistor | when Q-point precision is everything        |
| b) Firm biasing  | $R_{Th} \leq 0.1 \cdot \beta R_E$  | $R_E \geq 10 \cdot R_{Th} / \beta$  | $\leq \sim 9\text{--}10\%$ — acceptable in amplifiers             | • <b>large input resistance</b> — see below |



Loading error  $R_{Th}/(R_{Th}+\beta R_E)$  of the divider vs  $R_{Th}/(\beta R_E)$ . Malvino's two design points: stiff at 0.01 ( $\approx 1\%$  error) and firm at 0.1 ( $\approx 9\%$  error).

**Why does "firm" exist at all?** Stiff forces  $R_1$ ,  $R_2$  to be  $\sim 10\times$  smaller. Two costs: (i) the divider burns  $\sim 10\times$  more supply current; (ii) in an amplifier,  $R_1 \parallel R_2$  appears **in parallel with the signal input**, so a stiff divider drags the amplifier's input resistance down and loads the source. Firm biasing is the standard compromise:  $\sim 10\%$  Q-point tolerance bought back as **large input resistance** and low drain.

**HOT** If a PYQ says "design a  $\beta$ -independent amplifier," firm is the intended level; if it says "stiff," apply the 0.01 rule and say so.

**KEY Reconciling with Floyd-Boylestad (the earlier notes):** their rule " $R_2 \leq \beta R_E / 10$ " is (since  $R_{Th} < R_2$ ) roughly  $R_{Th} \leq 0.1 \beta R_E$  — i.e. Boylestad's "stiff" sits at Malvino's **firm** level. Numerically both conventions land on the same resistors (Design M1 below shows it). In the answer sheet, write the inequality you are using — that sentence is the method mark.

## D. • Design guidelines — Malvino splits by supply voltage

The feedback only works if  $V_E$  is **large compared to the  $\pm 0.1$  V unit-to-unit spread and the  $-2$  mV/ $^\circ\text{C}$  drift of  $V_{BE}$**  (otherwise  $V_{BE}$  noise eats the budget), yet **small compared to  $V_{CC}$**  (otherwise it eats output swing). How much room you have depends on  $V_{CC}$ :

★ For  $V_{CC} \geq 15$  V — the one-tenth rule:

$V_E = 0.1 \cdot V_{CC}$  — at  $V_{CC} = 15\text{ V}$  this is already  $1.5\text{ V} \gg 0.1\text{ V}$  of  $V_{BE}$  uncertainty.

$V(R_C) = 0.4 \cdot V_{CC}$ , i.e.  $R_C = 4 \cdot R_E$  (same current)  $\Rightarrow V_{CE} = V_{CC} - 0.4V_{CC} - 0.1V_{CC} = 0.5 \cdot V_{CC}$  — Q lands mid-load-line for maximum symmetric swing.

Divider:  $V_B = 0.1V_{CC} + 0.7 \ll V_{CC}$ , so  $R_2/(R_1+R_2)$  is small  $\Rightarrow R_1 \gg R_2 \Rightarrow R_{Th} = R_1 \parallel R_2 \approx R_2$  — you may apply the stiff/firm test directly to  $R_2$ .

★ **For  $V_{CC} \leq 10\text{ V}$  — the one-third rule:**

One-tenth would give  $V_E \leq 1\text{ V}$  — too close to the  $V_{BE}$  uncertainty. Spend a bigger fraction of the (small) supply on stability:

$V_{Th} = V_B = \frac{1}{3} \cdot V_{CC} \Rightarrow$  divider ratio  $R_2/(R_1+R_2) = 1/3 \Rightarrow R_1 = 2 \cdot R_2$  (exactly)  $\Rightarrow R_{Th} = R_1 \parallel R_2 = (2R_2 \cdot R_2)/(3R_2) = \frac{2}{3} \cdot R_2$

$V_{CE} \approx \frac{1}{3} \cdot V_{CC}$  as the Q-point target; then  $V_E = \frac{1}{3}V_{CC} - 0.7$  and  $V(R_C) = V_{CC} - V_C = \frac{1}{3}V_{CC} + 0.7$  — the supply is split into rough thirds.

(For  $10\text{ V} < V_{CC} < 15\text{ V}$  either guideline is defensible — the IOE PYQ solutions use the one-tenth rule, e.g.  $V_{CC} = 12\text{ V} \rightarrow V_E = 1.2\text{ V}$ .)

**KEY** Notice the division of labour: the voltage guideline fixes the divider's **ratio** (what  $V_B$  must be); the stiff/firm rule fixes the divider's **size** (how small  $R_{Th}$  must be). Two independent knobs  $\rightarrow R_1$  and  $R_2$  are fully determined.

**The recipe (memorize, write as sentences before substituting):**

1.  $V_E = 0.1V_{CC}$  (if  $V_{CC} \geq 15\text{ V}$ ) or from  $V_B = \frac{1}{3}V_{CC}$  (if  $V_{CC} \leq 10\text{ V}$ )  $\rightarrow R_E = V_E/I_E$ ,  $I_E = I_C(\beta+1)/\beta \approx I_C$ .
2.  $R_C = 4R_E$  (one-tenth rule) or from  $V_{CE} \approx \frac{1}{3}V_{CC}$  (one-third rule):  $R_C = (V_{CC} - V_E - V_{CE})/I_C$ . (CC design: no  $R_C$ ,  $V_E = V_{CC}/2$  instead.)
3.  $V_B = V_E + 0.7\text{ V} \rightarrow$  required ratio  $R_1/R_2 = (V_{CC} - V_B)/V_B$ . (One-third rule: ratio already fixed,  $R_1 = 2R_2$ .)
4. **Stiffness level:** stiff  $\rightarrow R_{Th} \leq 0.01\beta R_E$ ; firm  $\rightarrow R_{Th} \leq 0.1\beta R_E$ . Convert to  $R_2$  ( $R_{Th} \approx R_2$  for one-tenth designs;  $R_{Th} = \frac{2}{3}R_2$  for one-third designs) and pick the nearest standard value **below** the limit.
5.  $R_1$  from the ratio in step 3.
6. **Verify:** compute  $R_{Th}$  and check the inequality; optionally recompute  $I_E$  from the exact formula to state the residual error.

**E. HOT Worked designs — both guidelines, both stiffness levels**

**Design M1 (one-tenth rule) — PYQ 2081 Ashwin Q1 [4], redone Malvino-style: CE amplifier,  $V_{CC} = 20\text{ V}$ ,  $I_C = 1.5\text{ mA}$ ,  $\beta = 110$ .**

1.  $V_{CC} \geq 15\text{ V} \rightarrow V_E = 0.1 \times 20 = 2\text{ V}$ ;  $I_E = 1.5\text{ mA} \times 111/110 \approx 1.51\text{ mA} \rightarrow R_E = 2/1.51\text{ mA} \approx 1.32\text{ k}\Omega$  (1.3 k $\Omega$  std)
2.  $R_C = 4R_E \approx 5.3\text{ k}\Omega$  (check:  $V(R_C) = 0.4 \times 20 = 8\text{ V} \rightarrow R_C = 8/1.5\text{ mA} = 5.33\text{ k}\Omega$  ✓;  $V_{CE} = 10\text{ V}$ , mid-line)
3.  $V_B = 2.7\text{ V} \rightarrow$  ratio  $R_1/R_2 = (20 - 2.7)/2.7 = 6.41$ ;  $\beta R_E = 110 \times 1.32\text{ k} \approx 145\text{ k}\Omega$ ;  $R_{Th} = R_2 \parallel R_1 = 0.865 \cdot R_2$
4. **Firm:**  $R_{Th} \leq 0.1 \times 145\text{ k} = 14.5\text{ k}\Omega \rightarrow R_2 \leq 16.8\text{ k}\Omega \rightarrow R_2 = 15\text{ k}\Omega$ ,  $R_1 = 96\text{ k}\Omega$  (100 k $\Omega$  std). Check  $R_{Th} = 15\text{ k} \parallel 96\text{ k} \approx 13.0\text{ k}\Omega \leq 14.5\text{ k}\Omega$  ✓ (error  $\approx 9\%$ ).
5. **Stiff:**  $R_{Th} \leq 1.45\text{ k}\Omega \rightarrow R_2 \leq 1.68\text{ k}\Omega \rightarrow R_2 = 1.5\text{ k}\Omega$ ,  $R_1 = 9.6\text{ k}\Omega$  (10 k $\Omega$  std). Check  $R_{Th} \approx 1.3\text{ k}\Omega$  ✓ (error  $\approx 0.9\%$ ).

**KEY** Compare: the firm answer (15 k $\Omega$  / 96 k $\Omega$ ) is essentially the Boylestad-rule answer from the original notes (14.6 k $\Omega$  / 93.5 k $\Omega$ ) — the two conventions meet at firm. The stiff answer costs a divider ten times heavier: fine for a bias-precision question, poor for an amplifier front-end (input resistance collapses toward  $R_1 \parallel R_2 \approx 1.3\text{ k}\Omega$ ).

**Design M2 (one-third rule) — fresh numbers:**  $V_{CC} = 9\text{ V}$ ,  $I_C = 2\text{ mA}$ ,  $\beta = 100$ , *firm biasing*.

1.  $V_{CC} \leq 10\text{ V} \rightarrow V_{Th} = V_B = 9/3 = 3\text{ V} \rightarrow V_E = 3 - 0.7 = 2.3\text{ V}$ ;  $I_E = 2\text{ mA} \times 101/100 = 2.02\text{ mA} \rightarrow R_E = 2.3/2.02\text{ mA} \approx 1.14\text{ k}\Omega$  (1.2 k $\Omega$  std)
2.  $V_{CE} = 9/3 = 3\text{ V} \rightarrow V_C = V_E + V_{CE} = 5.3\text{ V} \rightarrow R_C = (9 - 5.3)/2\text{ mA} = 1.85\text{ k}\Omega$  (1.8 k $\Omega$  std)
3. Divider ratio fixed by the rule:  $R_1 = 2R_2$ ,  $R_{Th} = \frac{2}{3}R_2$ ;  $\beta R_E = 100 \times 1.14\text{ k} = 114\text{ k}\Omega$
4. **Firm:**  $\frac{2}{3}R_2 \leq 0.1 \times 114\text{ k} = 11.4\text{ k}\Omega \rightarrow R_2 \leq 17.1\text{ k}\Omega \rightarrow R_2 = 15\text{ k}\Omega$ ,  $R_1 = 30\text{ k}\Omega$  ( $R_{Th} = 10\text{ k}\Omega$  ✓)
5. **Verify with the exact formula** — the honest last line examiners like. **(a)** Firm ( $R_{Th} = 10\text{ k}\Omega$ ):  $I_E = (3 - 0.7)/(1.14\text{ k} + 10\text{ k}/101) = 2.3/1.239\text{ k} \approx 1.86\text{ mA}$  — 8% below target, exactly the firm-level tolerance predicted in §C. **(b)** Stiff option ( $R_2 = 1.5\text{ k}\Omega$ ,  $R_1 = 3\text{ k}\Omega$ ,  $R_{Th} = 1\text{ k}\Omega$ ):  $I_E = 2.3/(1.14\text{ k} + 9.9) \approx 2.00\text{ mA}$  — within 1%. ✓

**KEY** If the spec demands  $I_C = 2\text{ mA}$  *exactly* under firm biasing, absorb the loading term: choose  $R_E = (V_{Th} - 0.7)/I_E - R_{Th}/(\beta + 1) \approx 1.14\text{ k} - 0.10\text{ k} \approx 1.04\text{ k}\Omega$ . Saying this earns the "understands the approximation" mark.

## F. Common mistakes (Malvino edition) ●

- Applying the stiff test to  $R_2$  in a **one-third design** — there  $R_{Th} = \frac{2}{3}R_2$ , not  $\approx R_2$  (that shortcut is only for one-tenth designs where  $R_1 \gg R_2$ ).
- Calling  $R_{Th} \leq 0.1\beta R_E$  "stiff" — in Malvino's naming that is **firm**; stiff is 0.01.
- Using the one-tenth rule at  $V_{CC} = 5\text{--}9\text{ V}$  and ending up with  $V_E < 1\text{ V}$  — the design drifts with  $V_{BE}$ ; switch to the one-third rule.
- Computing  $R_1$  before  $R_2$  — impossible: **stiffness sets  $R_2$  (the size), the ratio then gives  $R_1$** .
- Forgetting that both rules assume  $I_E \approx I_C$ ; state it once.
- Quoting "negative feedback" without the chain — name every link:  $I_C \uparrow \rightarrow V_E \uparrow \rightarrow V_{BE} \downarrow \rightarrow \text{injection} \downarrow \rightarrow I_C$  pulled back.

## FORMULAS Formula Box — §1.6 (Malvino)

| Formula                                                                                                                                       | Role                               |
|-----------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|
| $V_{Th} = V_{CC} \cdot R_2 / (R_1 + R_2)$ , $R_{Th} = R_1 \parallel R_2$                                                                      | Thévenin base network              |
| $I_E = (V_{Th} - V_{BE}) / [R_E + R_{Th} / (\beta + 1)]$                                                                                      | exact bias current                 |
| $R_{in}(\text{base}) \approx \beta R_E$ ; error $\approx R_{Th} / \beta R_E$                                                                  | loading of the divider             |
| <b>Stiff:</b> $R_{Th} \leq 0.01\beta R_E$ ( $\Leftrightarrow R_E \geq 100R_{Th}/\beta$ )                                                      | ~1% Q-point error                  |
| <b>Firm:</b> $R_{Th} \leq 0.1\beta R_E$                                                                                                       | ~10% error, large input resistance |
| $V_{CC} \geq 15\text{ V}$ : $V_E = 0.1V_{CC}$ , $R_C = 4R_E$ , $V_{CE} = 0.5V_{CC}$ , $R_{Th} \approx R_2$                                    | one-tenth design                   |
| $V_{CC} \leq 10\text{ V}$ : $V_B = \frac{1}{3}V_{CC} \Rightarrow R_1 = 2R_2$ , $R_{Th} = \frac{2}{3}R_2$ ; $V_{CE} \approx \frac{1}{3}V_{CC}$ | one-third design                   |